

**Stable marriage (matching) problem**  
**Gale-Shapely algorithm**

1. What happens in GS if the last girl without a proposal gets her first proposal?
2. Show, that there is no boy-optimal matching with 2 boys forming a couple with the last girl on their list.
3. Show, that if 2 different girls are forming a couple with the last boy on their list in a boy-optimal matching, then there are at least 2 different stable matchings.